

Dhruv D. Bhrasadiya

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Introduction

Computer Engineering undergraduate focused on Computer Vision and deep learning research, with experience in analyzing research papers, benchmarking models, and evaluating performance on standard datasets. Worked on real-time vision systems and model optimization for edge deployment

Technical Skills

Languages: Python, JavaScript (ES6+), Java, C++, C

Frameworks & Libraries: React Native, FastAPI, Streamlit, Gradio, Redux, Node.js, Postman, Expo

ML/DL: PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face, OpenCV, Mediapipe, TensorRT, ONNX, Timm, Docker

Concepts: Computer Vision, Recommendation Systems, Deep Learning, API Integration, Mobile UI/UX, Model Optimization

Education

Vidyalankar Institute of Technology, B.Tech in Computer Science

Sept 2023 – May 2027

- GPA: 8.5/10.0
- **Coursework:** Machine Learning, Deep Learning, Computer Vision, Artificial Intelligence, Robotics, Probability and Statistics, Linear Algebra, Data Structures and Algorithms, Database Management Systems

Experience

Research Intern, Veermata Jijabai Technological Institute(VJTI) – Mumbai

May 2025 – July 2025

- Surveyed and analyzed research literature across Object Detection, Recognition, Classification & Segmentation for Vision Guided Systems, identifying key limitations and design trade-offs
- Conducted experiments across multiple model architectures, comparing performance and extracting insights to guide model selection and improvements
- Optimized selected models using TensorRT, achieving 30–40% latency reduction and sub-50ms inference on Jetson devices

Projects

Depth Nav

[Github](#)

- Developed a depth-aware navigation system operating at 30–40 FPS, enabling real-time directional adjustments for collision avoidance
- Implemented and compared fuzzy logic and rule-based decision strategies, reducing reliance on LiDAR and enabling low-cost edge deployment

Trash Detect

[Github](#)

- Optimized 8+ vision models using TensorRT/ONNX for real-time edge deployment, improving inference speed on resource-constrained devices.
- Developed a FastAPI backend and inference pipeline to deliver low-latency predictions via REST APIs for mobile/web integration.

AI-Powered Cricket Analytics

[Github](#)

- Developed a computer vision pipeline for analyzing 6+ batting shots using pose estimation and video processing, supporting video input from multiple camera angles
- Extracted biomechanical features from player movements to classify shot types and generate insights on shot quality and consistency across sequences

Certifications

NPTEL (IIT Kharagpur)

[NPTEL](#)

- Secured 71% in a nationally recognized course covering core DSA concepts

Achievements

LeetCode

[Leetcode](#)

- Solved 500+ problems with a 190+ day consistent streak, demonstrating strong algorithmic thinking and discipline